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A comprehensive comparative analysis of machine learning algorithms in heart disease prediction

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Abstract

This study investigates advanced machine learning techniques for predicting heart disease, emphasizing the critical role of early diagnosis in cardiovascular diseases (CVDs), which remain among the leading causes of mortality worldwide. Early and accurate detection can substantially reduce mortality rates and improve public health outcomes. In this context, advanced machine learning algorithms, such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest (RF), have emerged as powerful tools for analyzing complex patterns in medical data. The performance and accuracy of 13 different machine learning algorithms were evaluated in a binary classification task aimed at distinguishing between healthy individuals and those at high risk for heart disease. The combined dataset consisted of 1328 samples, with 50.08% classified as high-risk patients and 49.92% as healthy individuals, providing a balanced distribution for effective machine learning analysis. The dataset included 14 features, such as age, gender, blood pressure, cholesterol, and other health-related factors. Model performance was assessed using 13 different evaluation metrics, and results were reported as mean (m), standard deviation (SD), and root mean square error (RMSE). Pairwise comparisons of algorithms based on accuracy were performed using Significance Testing Between Models to evaluate statistically significant differences. Additionally, an exploratory data analysis was conducted to assess the influence of individual features on model outputs. The findings indicate that the RF algorithm achieved high accuracy ($93.82 \pm 1.64\%$) as well as high sensitivity, while models such as Extreme Gradient Boosting (XGB), Light Gradient Boosting Machine (LGBM), Extra Trees Classifier (ETC), and Decision Tree (DT) consistently ranked second and third across evaluation metrics. These results demonstrate that leveraging machine learning techniques enhances diagnostic accuracy, facilitates rapid identification of high-risk individuals, and reduces healthcare costs. Ultimately, this study provides a foundation for developing innovative prediction methods and management strategies for cardiovascular diseases.

Keywords Cardiovascular diseases (CVDs), Heart disease prediction, Machine learning, Classification algorithms

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1 Introduction

Cardiovascular diseases (CVDs) are recognized as one of the leading causes of mortality worldwide, and each year millions of people are dead due to CVs. These diseases account for over 70% of total deaths and significantly high economic costs impose on societies. Early diagnosis of heart disease can play a crucial role in reducing mortality rates and improving public health. Due to complexity and diversity of risk factors associated with heart diseases, applying machine learning (ML) algorithms for prediction and diagnosis of heart diseases can improve public health. [1, 2].

Machine learning, as one of the most advanced data analysis methods, can identify complex patterns in medical data, help physicians to make accurate predictions about the occurrence of cardiovascular diseases [1]. Various algorithms, such as Logistic Regression (LR), K-Nearest Neighbors (KNN), and Random Forest (RF), have significantly improved the accuracy and efficiency of prediction systems in medical data analysis. These algorithms, by precisely analyzing data and uncovering hidden patterns, contribute to faster and more accurate diagnosis of heart diseases, ultimately reducing treatment costs and early detection of heart disease [3].

Due to large volume and inherent complexity of medical datasets, advanced data analysis techniques such as machine learning algorithms are required [1]. Predicting heart diseases is one of the major challenges in public health, and by leveraging these techniques, high accuracy in identifying at-risk individuals can be achieved [4]. Additionally, due to the economic constraints and limited access to advanced diagnostic technologies such as electrocardiograms (ECG) and computed tomography scans (CT Scan), early prediction of heart diseases can reduce the financial costs of patients [2, 5, 6].

Considering the substantial impact of heart diseases on public health and their increasing prevalence due to lifestyle changes and unhealthy dietary habits, using machine learning techniques to accurately predict these diseases can enhance diagnostic precision [7–9]. This study presents a comprehensive and analytical investigation of advanced machine learning techniques for heart disease prediction, aiming to improve the accurate identification of high-risk individuals, enhance diagnostic precision, and reduce the costs associated with treatment and early detection. The analysis is conducted using integrated datasets formed by merging two heart disease data sources to improve data diversity and robustness. A systematic comparative evaluation of various classification algorithms is performed, focusing not only on performance assessment using multiple evaluation metrics but also on in-depth analytical comparisons between models. The most effective algorithms are identified based on their predictive performance, and statistical significance tests are applied to examine whether the observed performance differences among algorithms are meaningful and reliable. The findings highlight the potential of machine learning-based approaches in addressing the critical public health challenge posed by heart diseases and emphasize the importance of timely and accurate diagnosis.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive review of related studies in this field. Section 3 describes the methodology and the integrated datasets used in this research. Section 4 introduces the machine learning algorithms considered. Section 5 presents the evaluation metrics along with their mathematical formulations. Section 6 reports the experimental results and comparative

performance analysis. Finally, Sect. 7 concludes the paper and discusses future research directions.

2 Literature review

In recent years, numerous machine learning approaches have been applied to forecast cardiovascular diseases. For example, Vikas Chaurasia et al. [10] created a system for early prediction that incorporated techniques like Classification and Regression Trees (CART), Decision Tables (DT), Iterative Partitioning Techniques (IPT), tenfold cross-validation (10-FCV), and algorithms such as Naïve Bayes (NB), J48 Decision Trees, and Bagging. Their results showed that the Bagging technique surpassed the others, reaching an accuracy level of 83%. Moreover, Sani et al. [11] presented a method relying on Rule-Based Classifiers (RBC) and Sequential Minimal Optimization (SMO) to identify coronary artery disease. This technique successfully pinpointed major risk factors for heart diseases, although separating heart diseases from other health conditions continued to pose a considerable difficulty.

In a similar vein, Rao et al. [12] employed machine learning strategies to detect risk factors for heart disease, including hypertension, diabetes, smoking, and elevated cholesterol. Despite these progressions, precisely distinguishing between cardiovascular diseases and other medical conditions persists as a key obstacle. The intensity of heart diseases has also been evaluated through data mining (DM) and neural networks (NN). Coronary heart disease (CHD), as one of the most intricate medical issues, demands exceptional accuracy in examination. Delayed identification of CHD may lead to serious heart-related issues or fatality.

For building a predictive framework for heart disease (HD), Mohan et al. [13] suggested a combined machine learning strategy. They further proposed an innovative method for deriving essential features from data to train and assess machine learning classifiers (MLC), attaining a classification accuracy of 88%.

Olaniyan et al. [14] devised a three-phase approach for forecasting heart disease in individuals suffering from angina pectoris. Their method, which employed an Artificial Neural Network (ANN), attained an accuracy of 89%. Additionally, Ratnasari et al. [15] applied a feature reduction method for X-ray images based on Principal Component Analysis (PCA) and Region of Interest (ROI) with a gray-level threshold of 150. The majority of earlier studies utilized datasets containing 13 features, typically applying classification techniques to ascertain the presence or absence of heart disease. The Cleveland dataset has served as the main data source in these investigations.

Ahmad et al. [1] explored machine learning algorithms, including Logistic Regression, K-Nearest Neighbors, Support Vector Machine, and Gradient Boosting Classifier, for heart disease prediction. Their study showed that Extreme Gradient Boosting (XGBoost) with GridSearchCV hyperparameter optimization achieved the highest performance, with 100% training and 99% testing accuracy across datasets. They emphasized hyperparameter optimization's role in enhancing model effectiveness and recommended combining Feature Selection with GridSearchCV in future research, advancing diagnostic accuracy and risk factor identification.

Table 1 Machine learning algorithms which are used in this paper

1	**Adaptive Boosting**
2	Decision Tree Classifier
3	Extra Trees Classifier
4	K-Nearest Neighbors
5	Light Gradient Boosting Machine
6	Linear Discriminant Analysis
7	Logistic Regression
8	Multilayer Perceptron
9	Quadratic Discriminant Analysis
10	Random Forest Classifier
11	Support Vector Machine
12	Voting Ensemble based on Naive Bayes
13	Extreme Gradient Boosting

Table 2 Evaluation metrics

1	**Accuracy (ACC)**
2	Precision (PRE)
3	Sensitivity (SEN)
4	Specificity (SPC)
5	F1 Score (F1)
6	False Positive Rate (FPR)
7	False Negative Rate (FNR)
8	Matthews Correlation Coefficient (MCC)
9	Fowlkes-Mallow (F-M)
10	Balanced Accuracy (BAL-ACC)
11	Brier Score
12	Cohen's Kappa (K)
13	Geometric Mean (G-mean)

3 Materials and methods

This section presents the materials and methods employed in this study, describing the machine learning models used for heart disease prediction, the integrated dataset composed of merged health-related data sources, and the associated clinical and demographic features. It also details the data preprocessing pipeline, which includes physiological preprocessing and outlier removal performed together, feature standardization, and categorical variable encoding. Furthermore, the evaluation framework is outlined, including the performance metrics and cross-validation strategies adopted to ensure a reliable and unbiased assessment of model performance.

3.1 Proposed method

The main contribution of this study is the development of a heart disease prediction system based on machine learning models. All classification algorithms listed in Table 1 are trained and evaluated using the merged dataset. It should be noted that some algorithms do not require a specific training phase, as explained in Sect. 4.

The performance of these models was assessed using various evaluation metrics outlined in Table 2.

Previous studies on heart disease prediction were compared with the findings of this research. The main objective of this study is to broaden the application of machine learning models for heart disease prediction by considering a wide range of algorithms

and performing a comprehensive evaluation of their performance. This research utilizes an integrated dataset constructed by merging two publicly available heart disease datasets from Kaggle, namely the Heart Disease Cleveland dataset containing 303 samples [16] and the Heart Disease dataset containing 1025 samples [17]. The combined dataset includes health related attributes such as age, gender, blood pressure, cholesterol level, blood sugar levels, exercise induced conditions, family history of heart disease, and other relevant clinical factors. These data are used to train multiple machine learning models and to assess their predictive performance in the context of heart disease prediction. The overall proposed methodology of the study is illustrated in Fig. 1.

3.2 Dataset description

The heart disease prediction dataset used in this paper is an integrated dataset constructed by merging two publicly available heart disease datasets. It consists of 14 columns, including 13 independent features and 1 class label. The features of this dataset include age, gender, blood pressure, cholesterol level, blood sugar level, exercise status, family history of heart disease, and several other health-related factors. The target variable is divided into two groups: individuals with heart disease and those without it. Table 3 presents a structured summary of the integrated dataset and its features. In this article, this table has been compiled and organized based on the characteristics of the merged dataset to provide a clear and systematic representation of the data.

This integrated dataset, constructed by merging two publicly available heart disease datasets, includes information about factors that can influence the risk of heart disease. Figure 2 presents the heatmap of the dataset features, generated in this article to visualize the relationships between them. Correlation values range from −1 to 1, where positive values indicate a direct relationship, negative values indicate an inverse relationship, and values close to zero suggest little to no correlation.

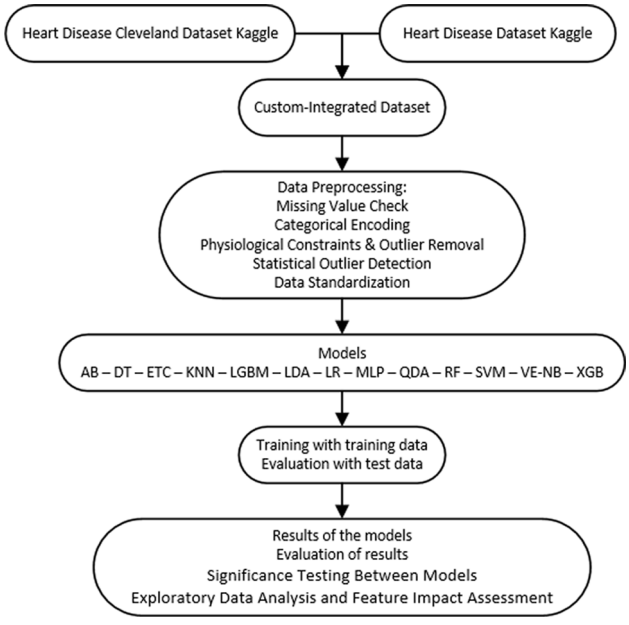


Fig. 1 Block Diagram of the process of using Machine Learning Techniques for heart disease prediction

Table 3 Description of attributes and class labels of the integrated heart disease dataset

Sr. no	Feature	Characteristic representation	Specifics
1	Age	age	Patients' age, in years
2	Sex	sex	0 = Female, 1 = Male
3	Chest pain	cp	4 types of chest pain (0-typical angina, 1-atypical angina, 2-non-anginal pain, 4-asymptomatic
4	Rest blood pressure	trtbps	Blood pressure normal (in mm Hg on admission to the hospital)
5	Serum cholesterol	chol	Cholesterol level in mg/dl
6	Fasting blood sugar	fbs	Fasting blood sugar > 120mg/dl (0-false, 1-true)
7	Rest electrocardiograph	restecg	0-normal, 1-having ST-T wave abnormality, 2-left ventricular hypertrophy
8	Max Heart rate	thalachh	Heart rate achieved which is at its maximum value
9	Exercise-induced angina	exng	Exercise-induced angina (0-no, 1-yes)
10	ST depression	oldpeak	ST depression induced by exercise relative to rest
11	Slope	slp	The slope of the peak exercise ST segment (0-upsloping, 1-fat, 2-down sloping)
12	No. of vessels	caa	No. of major vessels (0–4) colored by fluoroscopy
13	Thalassemia	thall	Type of thalassemia (0-Special condition, 1-Normal, 2-Fixed defect, 3-Reversible defect)
14	Num (class attribute)	output	Class label (0-Not sick, 1-Sick)

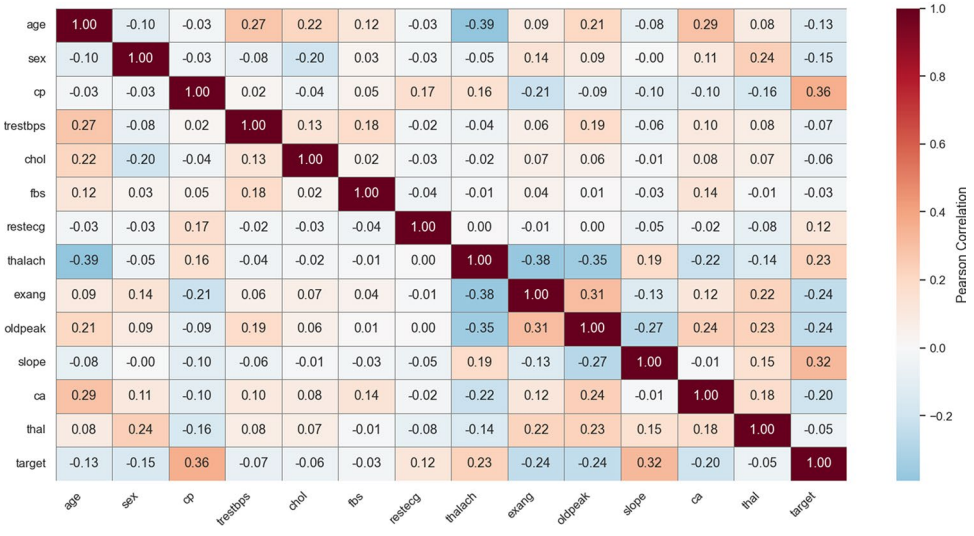


Fig. 2 Pearson correlation matrix between various features in predicting heart disease. Red indicates a strong positive correlation, blue indicates a strong negative correlation, and white indicates no correlation

The integrated dataset used in this study contains 1328 samples. The class distribution indicates that 50.08% of the samples belong to the heart disease class (target = 1), while 49.92% correspond to the non-disease class (target = 0).

In this paper, for better understanding the various features of the data, a set of descriptive statistics for the integrated dataset has been examined. These descriptive statistics include values such as mean (mean), standard deviation (std), minimum (min), 25th percentile (25%), median (50%), 75th percentile (75%), and maximum (max) for each feature in the dataset. Table 4 presents the descriptive statistics of the merged heart disease dataset before preprocessing.

Table 4 Descriptive statistics of the features in the integrated heart disease dataset samples

Feature	Mean	Std	Min	25%	50%	75%	Max
Age	54.43	9.06	29	48	56	61	77
Sex	0.69	0.46	0	0	1	1	1
Chest pain (Cp)	1.22	1.13	0	0	1	2	3
Rest blood pressure (Trtbps)	131.63	17.53	94	120	130	140	200
Serum cholesterol (Chol)	246.16	51.62	126	211	240	275	564
Fasting blood sugar (Fbs)	0.15	0.36	0	0	0	0	1
Rest electrocardiograph (Restecg)	0.63	0.69	0	0	1	1	2
Max Heart rate (Thalachh)	149.23	22.97	71	132	152	166	202
Exercise-induced angina (Exng)	0.33	0.47	0	0	0	1	1
ST depression (Oldpeak)	1.06	1.17	0	0	0.8	1.8	6.2
Slope (Slp)	1.21	0.7	0	1	1	2	2
No. of vessels (Caa)	0.73	1.01	0	0	0	1	4
Thalassemia (Thall)	2.21	0.74	0	2	2	3	3

3.3 Hardware specification and data preprocessing

The experiments in this study were conducted on a Lenovo laptop model 81Y3 running Windows 10 Enterprise version 10.0.19045, powered by an Intel Core i7 processor with × 64 architecture. Initially, the data were loaded and processed using Python version 3.13.0 installed on a 64-bit Windows platform. Several libraries were utilized for data preprocessing in this study. To ensure data accuracy and prepare the integrated dataset for modeling, a sequence of preprocessing steps was performed. In the first step, the integrated dataset was examined for missing values. No missing values were detected; therefore, no imputation techniques were required.

In the second step, categorical and non-numeric features were transformed into numerical representations based on the definitions provided in Table 3. This transformation ensured that all attributes were expressed in a consistent numerical format and were harmonized across the merged datasets, making them suitable for unified analysis and modeling within the integrated dataset.

In the third step, physiological preprocessing and outlier removal were jointly applied to identify and eliminate implausible and extreme values. Physiological constraints were first imposed on clinically meaningful features, including age, resting blood pressure, cholesterol level, maximum heart rate, and ST depression, to remove values outside realistic human ranges. Subsequently, statistical outlier detection was performed using the interquartile range method and Z-score analysis. Samples lying beyond 1.5 times the interquartile range or exceeding 3 standard deviations from the mean were excluded. Through this combined procedure, 83 anomalous samples were identified and removed. As a result, the cleaned integrated dataset consisted of 1245 samples. The class prevalence in the cleaned dataset was 50.92% for the heart disease class (target = 1) and 49.08% for the non-disease class (target = 0).

In the fourth step, due to differences in the scale of various numerical features, such as cholesterol level compared to age, data standardization was applied. The StandardScaler method from the Scikit Learn library was used to normalize the features according to Eq. (1), where X denotes the original feature value, μ represents the mean, and σ indicates the standard deviation. This transformation rescaled the data to have a zero mean and unit variance, thereby improving the stability and performance of the machine learning models.

$$\frac{(X-\mu)}{\sigma} \tag{1}$$

Fivefold stratified cross-validation was used to evaluate the models. In this method, the dataset is divided into 5 folds, ensuring that the class prevalence in each fold matches the overall dataset. Each fold is used once as the test set, and the results are averaged to obtain a stable performance estimate. All preprocessing steps described above, including data cleaning, outlier removal, and feature standardization, were applied to the fully integrated dataset prior to performing the fivefold stratified cross-validation.

4 Machine learning techniques for predicting heart diseases

In this section, a diverse set of machine learning models is employed for heart disease prediction, with each algorithm selected based on its distinct characteristics, underlying assumptions, and strengths in addressing different aspects of classification tasks. The model selection strategy was deliberately designed to enable a comprehensive and systematic comparison across heterogeneous learning paradigms, including linear and nonlinear classifiers, probabilistic and discriminative methods, instance-based learning approaches, neural networks, and ensemble techniques based on bagging, boosting, and voting mechanisms. This methodological diversity allows for an in-depth investigation of how factors such as linear separability, feature independence, class-conditional distributions, and model complexity influence predictive performance on a real-world clinical dataset. For all thirteen algorithms, hyperparameter optimization was conducted using a grid search strategy over predefined model-specific parameter spaces, combined with stratified five-fold cross-validation to ensure robust and unbiased evaluation. The F1-score was adopted as the optimization criterion to account for potential class imbalance and to achieve balanced performance across classes. Furthermore, uniform data preprocessing and consistent evaluation protocols were applied across all models to guarantee a fair and reproducible comparative analysis. The inclusion of both classical algorithms (e.g., Logistic Regression, LDA, QDA, and KNN) and state-of-the-art ensemble methods (e.g., Random Forest, Extra Trees, LightGBM, XGBoost, and AdaBoost) provides clinically meaningful insights into the trade-offs between interpretability, robustness, and predictive accuracy, thereby enhancing the reliability and applicability of the proposed heart disease risk prediction framework.

4.1 Adaptive boosting (AB) model

Adaptive Boosting (AdaBoost) is an ensemble boosting algorithm introduced by Freund and Schapire that constructs a strong classifier by iteratively combining multiple weak learners and adaptively reweighting misclassified samples [18]. In accordance with the implemented code, AdaBoost was optimized via a grid-search strategy in which the number of weak learners n_estimators $\in \{50, 100, 200\}$ and the learning rate $\in \{0.01, 0.1, 1\}$ were systematically explored. Each weak learner corresponds to a shallow decision tree, and the final prediction is obtained through a weighted majority vote of these learners. The reweighting mechanism increases the influence of misclassified instances at each boosting iteration, thereby minimizing the exponential loss. Based on cross-validation results, the optimal configuration was achieved with n_estimators = 200 and learning_rate = 0.1, which was subsequently used to train the final AdaBoost model.

4.2 Decision tree classifier (DT) model

Decision Tree is a supervised learning algorithm that recursively partitions the feature space into homogeneous regions by selecting splits that maximize class purity at each node [19]. In the present implementation, the classifier was optimized over the hyper-parameter space defined by max_depth ∈ {None, 5, 10, 15} and min_samples_split ∈ {2, 5, 10} while fixing random_state = 42 to ensure reproducibility. Node impurity was measured using the Gini index,

$$Gini = 1 - \sum_{k=1}^{K} P_k^2, \tag{2}$$

where P_k denotes the proportion of samples belonging to class *k*. The grid-search procedure identified the optimal tree with unlimited depth (max_depth = None) and min_samples_split = 2, allowing the model to fully adapt its structure to the underlying data distribution.

4.3 Extra trees classifier (ETC) model

Extra Trees Classifier is an ensemble of decision trees that extends the Random Forest paradigm by injecting additional randomness in both feature selection and split thresholds, thereby reducing variance and computational cost [20]. In the implemented framework, the model was tuned across n_estimators ∈ {200, 300, 500} and max_depth ∈ {None, 5, 10, 15}, with random_state = 42 fixed for consistency. Predictions are obtained through majority voting across trees. The optimal configuration, as determined by cross-validation, consisted of n_estimators = 300 and max_depth = 15, which provided the best balance between model complexity and generalization performance.

4.4 K-nearest neighbors (KNN) model

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm that assigns a class label to a query point based on the most frequent class among its *k* nearest neighbors in the feature space [21]. Distance computation relies on the Minkowski metric,

$$d_p\,(x.y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}, \tag{3}$$

which reduces to the Euclidean distance when p = 2. In the present study, the model was optimized over n_neighbors ∈ {3, 5, 7} and weights ∈ {uniform, distance}. The best-performing configuration employed n_neighbors = 7 with distance-based weighting, ensuring that closer neighbors exert greater influence on the classification outcome.

4.5 Light gradient boosting machine (LGBM) model

LightGBM is a gradient-boosting framework based on decision trees, distinguished by its leaf-wise tree growth strategy that accelerates convergence and improves accuracy on high-dimensional data [22]. In line with the code implementation, a grid search was conducted over n_estimators ∈ {200, 300, 500}, max_depth ∈ {3, 5, 7}, and learning_rate ∈ {0.01, 0.1}. Gradient boosting iteratively minimizes a differentiable loss function by fitting new trees to the negative gradients of the loss. The optimal hyper-parameter set

was found to be n_estimators = 500, max_depth = 7 and learning_rate = 0.01, which was adopted for the final LightGBM model.

4.6 Linear discriminant analysis (LDA) model

Linear Discriminant Analysis is a probabilistic classifier that seeks linear combinations of features maximizing the ratio of between-class scatter to within-class scatter, thereby enhancing class separability [23]. The transformation is obtained by maximizing

$$J(w) = \frac{w^T S_B w}{w^T S_W w}, \tag{4}$$

where S_B and S_W denote the between-class and within-class scatter matrices, respectively. In this study, LDA was optimized over solver ∈ {svd, lsqr, eigen} and shrinkage ∈ {None, auto}. Cross-validation results indicated that the optimal configuration employed the SVD solver with no shrinkage (solver = svd, shrinkage = None).

4.7 Logistic regression (LR) model

Logistic Regression is a linear probabilistic classifier that models the posterior probability of a binary outcome using the logistic function [24],

$$P(y = 1|x) = \frac{1}{1 + e^{-w^T x}}, \tag{5}$$

In the implemented code, the model was tuned over the regularization strength C ∈ {0.1, 1, 10}, with an L_2 penalty and the LBFGS solver, while max_iter = 1000 ensured convergence. The optimal configuration was obtained for C = 0.1, penalty = l2, and solver = lbfgs.

4.8 Multilayer perceptron (MLP) model

Multilayer Perceptron is a feedforward neural network composed of interconnected layers of neurons capable of approximating nonlinear decision boundaries [25]. Each neuron applies a nonlinear activation function to a weighted sum of its inputs. In this work, the MLP was optimized with respect to hidden_layer_sizes ∈ {(50,), (100,)} and activation ∈ {relu, tanh}, with max_iter = 1000 and random_state = 42. The rectified linear unit,

$$ReLU(z) = \max(0.z), \tag{6}$$

yielded the best performance when combined with a single hidden layer of 100 neurons, which was therefore selected as the optimal configuration.

4.9 Quadratic discriminant analysis (QDA) model

Quadratic Discriminant Analysis is a generative classifier that models each class with its own Gaussian distribution and class-specific covariance matrix, resulting in quadratic decision boundaries [26]. The discriminant function depends on the class-conditional density,

$$p(x|y = k) = \frac{1}{(2\pi)^{d/2} \left|\sum_k\right|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \sum_k^{-1} (x - \mu_k)\right), \tag{7}$$

In the implemented grid search, the regularization parameter reg_param ∈ {0.0, 0.01, 0.05, 0.1} was varied to stabilize covariance estimation. The optimal model was obtained with reg_param = 0.1.

4.10 Random forest classifier (RF) model

Random Forest is an ensemble learning method that aggregates multiple decision trees trained on bootstrap samples and random feature subsets, thereby reducing variance and improving generalization [27]. In this study, the model was optimized over n_estimators ∈ {200, 300, 500}, max_depth ∈ {None, 5, 10, 15}, min_samples_split ∈ {2, 5, 10} and min_samples_leaf ∈ {1, 2, 4}, with random_state = 42. The final optimal configuration comprised n_estimators = 500, max_depth = None, min_samples_split = 2 and min_samples_leaf = 1.

4.11 Support vector machine (SVM) model

Support Vector Machine is a margin-based classifier that seeks the hyperplane maximizing the separation margin between classes [28]. In the nonlinear case, the optimization problem is solved in a transformed feature space induced by a kernel function. In the present work, SVM was tuned over C ∈ {0.1, 1, 10} and kernel ∈ {linear, rbf}, with probability estimates enabled. The radial basis function kernel,

$$K\left(x.x^{\cdot}\right) = \exp\left(-\gamma x - x^{\cdot 2}\right), \tag{8}$$

combined with C = 10, yielded the optimal classification performance.

4.12 Voting ensemble based on Naive Bayes model

The Voting Ensemble based on Naive Bayes integrates Gaussian Naive Bayes and Bernoulli Naive Bayes classifiers within a soft-voting framework, where class probabilities predicted by individual models are averaged to form the final decision [29, 30]. Gaussian Naive Bayes assumes continuous features following class-conditional Gaussian distributions, whereas Bernoulli Naive Bayes models binary-valued features. Under the conditional independence assumption, the posterior probability is computed via Bayes' theorem. In the implemented ensemble, hyper-parameters were optimized over gnb__var_smoothing ∈ $\{10^{-9}.10^{-8}.10^{-7}\}$ and bnb__alpha ∈ {0.1, 0.5, 1.0}. The optimal ensemble employed gnb__var_smoothing = 10^{-9} and bnb__alpha = 0.1 providing stable probabilistic predictions.

4.13 Extreme gradient boosting (XGB) model

Extreme Gradient Boosting is an advanced boosting algorithm that incrementally builds an ensemble of decision trees by minimizing a regularized objective function through gradient-based optimization [31]. In this study, the XGBoost classifier was optimized over n_estimators ∈ {200, 300, 500}, max_depth ∈ {3, 5, 7}, and learning_rate ∈ {0.01, 0.1, 0.2}, with eval_metric = logloss and random_state = 42. The best-performing configuration consisted of n_estimators = 500, max_depth = 7 and learning_rate = 0.01, which was subsequently adopted for final model training and evaluation.

5 Model evaluation metrics

In this study, the performance of prediction models was evaluated using a set of common metrics in machine learning. These metrics assist in assessing the accuracy of models in identifying heart diseases based on the available data [32, 33].

Before delving into the evaluation metrics, it is essential to understand the confusion matrix and its components. The confusion matrix is a table used to describe the performance of a classification model, where rows correspond to actual classes and columns represent predicted classes. For binary classification problems, like predicting heart disease (with two classes: healthy and diseased), the confusion matrix consists of the following four values:

- True Positive (TP): Instances where the model correctly classifies diseased individuals as diseased.
- True Negative (TN): Instances where the model correctly classifies healthy individuals as healthy.
- False Positive (FP): Instances where the model incorrectly classifies healthy individuals as diseased (Type I error).
- False Negative (FN): Instances where the model incorrectly classifies diseased individuals as healthy (Type II error).

The following confusion matrix table is proposed for defining the performance evaluation in this study:

Confusion matrix	Predicted disease (Class 1)	Predicted HEalthy (Class 2)
Actual disease (Class 1)	True Positive (TP)	False Negative (FN)
Actual healthy (Class 2)	False Positive (FP)	True Negative (TN)

Now, using these four values from the confusion matrix, we can calculate the following evaluation metrics [32, 33]:

Accuracy: The model's overall accuracy is calculated as the ratio of correct predictions to the total number of predictions. Its formula is:

$$Accuracy = \frac{(\mathrm{TP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})} \tag{9}$$

Reflects the overall correctness of the model, providing a general measure of its ability to classify patients correctly, which is crucial in clinical decision-making.

Precision: Also known as positive predictive value (PPV), precision indicates the model's ability to identify positive cases correctly and is calculated as:

$$Precision = \frac{TP}{(TP + FP)} \tag{10}$$

Indicates how reliably a positive prediction corresponds to actual disease, important to reduce unnecessary interventions for healthy patients.

Sensitivity (Recall): Also referred to as the true positive rate (TPR) or detection rate, sensitivity measures the model's ability to identify all positive instances. Its formula is:

$$Sensitivity = \frac{TP}{(TP + FN)} \tag{11}$$

Measures the model's ability to detect all diseased cases, essential for early diagnosis and minimizing missed cases.

Specificity: Specificity, also known as the true negative rate (TNR), measures the model's success in identifying negative cases and follows the formula:

$$Specificity = \frac{TN}{TN + FP} \tag{12}$$

Assesses the model's ability to correctly identify healthy individuals, helping to avoid over-treatment in clinical practice.

F1 Score: The F1 score represents the harmonic mean of precision and recall, summarizing both metrics into a single value. It is calculated as:

$$F1 = \frac{Precision \times Sensitivity}{Precision + Sensitivity} \times 2 \tag{13}$$

Balances precision and recall, offering a single metric that considers both false positives and false negatives, useful in scenarios with clinical risk.

False Positive Rate (FPR): FPR indicates the proportion of incorrectly predicted positive cases and is calculated using:

$$FPR = \frac{FP}{FP + TN} \tag{14}$$

Reflects the proportion of healthy patients incorrectly labeled as diseased; minimizing FPR reduces unnecessary anxiety and medical costs.

False Negative Rate (FNR): FNR measures the proportion of incorrectly predicted negative cases and is expressed as:

$$FNR = \frac{FN}{TP + FN} \tag{15}$$

Represents the proportion of diseased patients missed by the model; reducing FNR is critical to prevent undiagnosed disease progression.

Prevalence: Prevalence indicates the percentage of total positive samples in the dataset and is calculated as:

$$Prevalence = \frac{TP + FN}{TP + TN + FP + FN} \tag{16}$$

Indicates the proportion of disease in the dataset, providing context for interpreting other metrics in clinical populations.

Matthews Correlation Coefficient (MCC): The MCC is a comprehensive metric that considers all TP, TN, FP, and FN values for model evaluation. It is calculated as:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)\,(TP + FN)\,(TN + FP)\,(TN + FN)}} \tag{17}$$

Provides a balanced evaluation accounting for all four confusion matrix categories, valuable when classes are imbalanced.

Fowlkes-Mallows Index (F-M): The Fowlkes-Mallows score is calculated as the geometric mean of precision and recall and serves as a useful metric for model evaluation:

$$FM = \sqrt{\mathrm{Pr}\,ecision \times Sensitivity} \tag{18}$$

Evaluates the geometric mean of precision and recall, emphasizing balanced detection performance relevant for clinical screening.

Balanced Accuracy: Balanced Accuracy is the mean of recall and specificity and is calculated using the formula:

$$BalancedAccuracy = \frac{Sensitivity + Specificity}{2} \tag{19}$$

Combines sensitivity and specificity, ensuring fair evaluation in datasets with uneven class distribution.

Brier Score: The Brier score is another metric for evaluating model accuracy, measuring the difference between the model's predictions and actual outcomes:

$$BrierScore = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2 \approx \frac{FN + FP}{TP + TN + FP + FN} \tag{20}$$

f_i is the model's prediction, and o_{i} is the actual outcome. Measures the mean squared difference between predicted probabilities and actual outcomes, offering insight into the reliability of probabilistic predictions for risk stratification.

Cohen's Kappa (κ): Cohen's Kappa is a metric for measuring agreement between predictions and actual outcomes, calculated as:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{21}$$

P_o represents observed agreement and P_e represents chance agreement [34].

$$P_o = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

$$P_e = \frac{(TN + FP)(TN + FN) + (TP + FN)(TP + FP)}{(TP + TN + FP + FN)^2} \tag{23}$$

Quantifies agreement between predicted and actual classifications beyond chance, important for validating clinical prediction consistency.

G-Mean Score (Gmean): The G-Mean score is a balanced metric for evaluating the model's performance in identifying both positive and negative classes, calculated as:

$$Gmean = \sqrt{Sensitivity \times Specificity} \tag{24}$$

Reflects balanced performance across both positive and negative classes, emphasizing robust diagnostic ability.

In the evaluation of predictive models, it is essential to consider the nature of each metric. Specifically, for certain metrics, a lower value indicates better model performance. These metrics include FPR, FNR, and Brier Score. Lower values for these metrics signify fewer errors in predictions, meaning the model is making fewer false classifications. Therefore, minimizing these criteria is very important to improve the accuracy and reliability of the model. Conversely, for other metrics, higher values indicate better performance and are of high importance. Using these metrics, the performance of

various models was evaluated based on accuracy, sensitivity, precision, and other characteristics. We calculate the mean and standard deviation for each evaluation metric. Furthermore, to enhance the comparison, we also compute the Root Mean Square Error (RMSE).

Root Mean Square Error (RMSE): RMSE is a widely used metric for evaluating the difference between predicted and actual values. It measures the average magnitude of the errors in a set of predictions, with larger errors contributing more heavily to the score. It is calculated as follows and related to the mean (m) and standard deviation (sd) of the error:

$$RMSE = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(e_j)^2} = \sqrt{(m)^2 + (sd)^2} \tag{25}$$

where $e_j = \hat{y}_j - y_j$ and y_j represents the actual value, and $\hat{y}_j$ represents the predicted value for the j-th observation. RMSE is particularly useful when penalizing large errors is crucial in model evaluation. We present the results as the mean ± standard deviation, with the value inside the parentheses representing the RMSE.

Significance Testing Between Models: To evaluate whether differences in predictive performance among the algorithms are statistically meaningful, a pairwise significance test was conducted using the 60% overlap rule on the accuracy distributions. For each pair of models, the mean ± standard deviation of accuracy was compared, and the degree of overlap between intervals was calculated. If the overlap was less than 60% of the smaller interval, the difference was considered significant. This approach allowed us to identify models that perform meaningfully better than others while considering variability in performance due to data sampling. Results were summarized in a pairwise comparison table, indicating which model pairs exhibited significant differences, providing a rigorous and interpretable framework for algorithm selection in clinical prediction of heart disease.

6 Results

Using fivefold cross-validation on the integrated, preprocessed, and cleaned dataset—where Class 1 represents individuals at high risk of heart disease and Class 2 denotes healthy subjects—thirteen machine learning algorithms were systematically evaluated. Model performance was assessed using the mean (m), standard deviation (sd), and Root Mean Square Error (RMSE) to quantify both predictive accuracy and model stability. The comprehensive results of this evaluation are presented in Table 5, in which the top three performing algorithms for each metric are highlighted in bold to facilitate comparison. While the current study uses a moderate-sized integrated dataset, the computational approach and chosen algorithms are generally scalable, and the potential for real-time clinical deployment is discussed in terms of model efficiency and resource requirements.

The experimental results indicate that the Random Forest (RF) model demonstrates consistently strong performance across most evaluation criteria. In particular, RF achieved high values for key classification metrics such as accuracy (93.82 ± 1.64), precision (94.57 ± 1.24), and F1-score (93.86 ± 1.68), while also maintaining relatively low false positive and false negative rates. This balanced performance can be attributed to

Table 5 Performance results of machine learning algorithms based on evaluation metrics

Mean % ± standard deviation % (RMSE %)													
	AB	**DT**	**ETC**	**KNN**	**LGBM**	**LDA**	**LR**	**MLP**	**QDA**	**RF**	**SVM**	**VE-NB**	**XGB**
Accuracy	76.14 ± 2.47 (76.18)	92.05 ± 1.65 (92.06)	**93.49 ± 2.12 (93.52)**	91.65 ± 1.28 (91.66)	**93.33 ± 2.21 (93.36)**	73.25 ± 2.15 (73.28)	73.65 ± 1.56 (73.67)	92.93 ± 1.45 (92.94)	78.88 ± 0.74 (78.88)	**93.82 ± 1.64 (93.83)**	91.00 ± 1.98 (91.03)	70.68 ± 2.33 (70.72)	93.17 ± 1.61 (93.19)
Precision	75.85 ± 4.14 (75.96)	92.13 ± 1.64 (92.14)	**94.10 ± 1.42 (94.11)**	92.63 ± 1.41 (92.64)	**94.37 ± 1.50 (94.38)**	72.22 ± 1.48 (72.23)	72.99 ± 1.25 (73.00)	93.90 ± 1.34 (93.91)	77.51 ± 1.01 (77.52)	**94.57 ± 1.24 (94.58)**	91.70 ± 1.81 (91.72)	70.24 ± 2.25 (70.27)	94.07 ± 0.86 (94.08)
Sensitivity	78.71 ± 2.82 (78.76)	**92.27 ± 1.96 (92.29)**	**93.06 ± 3.71 (93.13)**	90.85 ± 2.52 (90.89)	92.43 ± 3.95 (92.51)	77.12 ± 4.45 (77.25)	76.65 ± 3.79 (76.74)	92.11 ± 2.17 (92.14)	82.49 ± 2.20 (82.52)	**93.22 ± 2.93 (93.26)**	90.53 ± 3.02 (90.58)	73.65 ± 3.06 (73.71)	92.43 ± 3.41 (92.49)
Specificity	73.51 ± 7.17 (73.86)	91.82 ± 1.70 (91.84)	93.95 ± 1.49 (93.96)	92.48 ± 1.57 (92.49)	**94.28 ± 1.53 (94.29)**	69.23 ± 2.18 (69.26)	70.53 ± 2.38 (70.57)	93.78 ± 1.43 (93.79)	75.12 ± 2.02 (75.15)	**94.44 ± 1.30 (94.45)**	91.49 ± 1.89 (91.51)	67.59 ± 2.92 (67.66)	**93.95 ± 0.98 (93.95)**
F1 Score	77.11 ± 1.51 (77.12)	92.19 ± 1.66 (92.21)	**93.54 ± 2.18 (93.57)**	91.71 ± 1.34 (91.72)	**93.35 ± 2.31 (93.37)**	74.54 ± 2.61 (74.58)	74.72 ± 2.00 (74.75)	92.99 ± 1.47 (93.00)	79.90 ± 0.86 (79.90)	**93.86 ± 1.68 (93.88)**	91.09 ± 2.07 (91.11)	71.88 ± 2.40 (71.92)	93.21 ± 1.71 (93.22)
FPR	26.49 ± 7.17 (27.44)	8.18 ± 1.70 (8.36)	**6.05 ± 1.49 (6.23)**	7.52 ± 1.57 (7.69)	**5.72 ± 1.53 (5.93)**	30.77 ± 2.18 (30.85)	29.47 ± 2.38 (29.56)	6.22 ± 1.43 (6.38)	24.88 ± 2.02 (24.96)	**5.56 ± 1.30 (5.71)**	8.51 ± 1.89 (8.72)	32.41 ± 2.92 (32.54)	6.05 ± 0.98 (6.13)
FNR	21.29 ± 2.82 (21.47)	**7.73 ± 1.96 (7.98)**	**6.94 ± 3.71 (7.87)**	9.15 ± 2.52 (9.49)	7.57 ± 3.95 (8.54)	22.88 ± 4.45 (23.31)	23.35 ± 3.79 (23.66)	7.89 ± 2.17 (8.18)	17.51 ± 2.20 (17.65)	**6.78 ± 2.93 (7.39)**	9.47 ± 3.02 (9.94)	26.35 ± 3.06 (26.53)	7.57 ± 3.41 (8.31)
MCC	52.50 ± 4.55 (52.70)	84.10 ± 3.30 (84.17)	**87.06 ± 4.18 (87.16)**	83.36 ± 2.53 (83.40)	**86.76 ± 4.37 (86.87)**	46.60 ± 4.36 (46.81)	47.37 ± 3.14 (47.47)	85.90 ± 2.88 (85.95)	57.86 ± 1.52 (57.88)	**87.69 ± 3.25 (87.75)**	82.05 ± 3.94 (82.15)	41.36 ± 4.64 (41.62)	86.43 ± 3.16 (86.48)
F-M	77.20 ± 1.39 (77.21)	92.20 ± 1.65 (92.21)	**93.56 ± 2.17 (93.59)**	91.72 ± 1.33 (91.73)	**93.37 ± 2.29 (93.40)**	74.60 ± 2.65 (74.65)	74.77 ± 2.03 (74.80)	93.00 ± 1.47 (93.01)	79.95 ± 0.89 (79.95)	**93.88 ± 1.68 (93.89)**	91.10 ± 2.06 (91.13)	71.91 ± 2.40 (71.95)	93.23 ± 1.70 (93.24)
Balanced Accuracy	76.11 ± 2.52 (76.15)	92.04 ± 1.65 (92.06)	**93.50 ± 2.09 (93.53)**	91.66 ± 1.26 (91.67)	**93.35 ± 2.18 (93.38)**	73.17 ± 2.12 (73.20)	73.59 ± 1.53 (73.61)	92.95 ± 1.44 (92.96)	78.81 ± 0.74 (78.81)	**93.83 ± 1.62 (93.84)**	91.01 ± 1.97 (91.03)	70.62 ± 2.33 (70.66)	93.19 ± 1.57 (93.20)
Brier Score	17.55 ± 0.23 (17.56)	7.95 ± 1.65 (8.12)	**4.45 ± 1.16 (4.60)**	6.10 ± 0.88 (6.16)	5.99 ± 1.97 (6.31)	17.19 ± 0.92 (17.21)	17.19 ± 0.93 (17.21)	6.38 ± 1.02 (6.46)	15.08 ± 0.85 (15.10)	**5.09 ± 1.28 (5.24)**	7.08 ± 1.39 (7.21)	20.10 ± 1.66 (20.16)	**5.72 ± 1.20 (5.85)**
Cohen's Kappa	52.24 ± 5.00 (52.48)	84.09 ± 3.31 (84.16)	**86.99 ± 4.22 (87.09)**	83.29 ± 2.55 (83.33)	**86.67 ± 4.41 (86.78)**	46.41 ± 4.27 (46.61)	47.23 ± 3.09 (47.34)	85.86 ± 2.90 (85.91)	57.68 ± 1.49 (57.70)	**87.63 ± 3.27 (87.69)**	82.01 ± 3.95 (82.10)	41.29 ± 4.66 (41.55)	86.35 ± 3.20 (86.41)
G-mean	75.90 ± 2.90 (75.96)	92.04 ± 1.65 (92.06)	**93.48 ± 2.10 (93.51)**	91.64 ± 1.27 (91.65)	**93.32 ± 2.19 (93.35)**	73.01 ± 2.05 (73.04)	73.48 ± 1.49 (73.49)	92.94 ± 1.45 (92.95)	78.69 ± 0.75 (78.70)	**93.81 ± 1.62 (93.83)**	91.00 ± 1.97 (91.02)	70.53 ± 2.35 (70.57)	93.16 ± 1.59 (93.18)

the ensemble nature of RF, which aggregates multiple decision trees and thereby reduces variance and improves generalization. Such characteristics are especially valuable in clinical diagnostic settings, where reliable identification of patients and robustness across different data splits are essential. Accordingly, RF appears to be a well-suited and dependable option for clinical decision-support systems, rather than a universally optimal solution.

The Extra Trees Classifier (ETC) also exhibited competitive behavior and showed particular strength in probabilistic prediction, achieving the most favorable Brier Score (4.45 ± 1.16). This suggests that ETC provides well-calibrated probability estimates, which are crucial for risk-based clinical decision-making. In addition, ETC frequently ranked close to RF in several performance measures, including accuracy (93.49 ± 2.12), F1-score, and balanced accuracy, while displaying stable performance across cross-validation folds. The use of increased randomness in feature selection and split thresholds likely contributes to its robustness and resistance to overfitting.

Light Gradient Boosting Machine (LGBM), as a gradient-boosting-based ensemble model, generally ranked among the top-performing methods and demonstrated a balanced trade-off between sensitivity and specificity. Its ability to model complex nonlinear interactions through iterative boosting enables effective learning from structured medical data, although its performance remained slightly below that of the strongest tree-based ensembles in this study. In contrast, linear classifiers such as Logistic Regression (LR) and Linear Discriminant Analysis (LDA), along with simpler probabilistic approaches such as the Naïve Bayes Ensemble, tended to yield weaker results—particularly with respect to error-sensitive and correlation-based metrics. This behavior reflects the inherent limitations of such models in capturing complex, nonlinear relationships that are common in clinical datasets.

Overall, the findings suggest that tree-based ensemble models, especially RF and ETC, provide more consistent and robust performance compared to other algorithmic categories examined in this study. Their capacity to handle feature interactions, mitigate noise, and maintain stability across validation folds makes them promising and trustworthy tools for supporting early detection and clinical decision-making in heart disease diagnosis, within the scope and constraints of the investigated dataset.

The observed performance differences among models such as SVM, KNN, and ETC can be partially explained by the intrinsic characteristics of the dataset. The presence of heterogeneous features, moderate sample size, and nonlinear relationships among clinical variables tends to favor models capable of capturing complex decision boundaries, such as tree-based ensembles. Distance-based methods like KNN are more sensitive to feature scaling and local data distributions, which may limit their effectiveness in high-dimensional clinical data. Similarly, the performance of SVM is influenced by kernel selection and parameter tuning, and linear assumptions may restrict its ability to fully model nonlinear patterns. These observations suggest that dataset structure, feature interactions, and nonlinearity play a significant role in determining the relative suitability of different machine learning algorithms for heart disease prediction.

Table 6 reports the results of the pairwise significance analysis conducted using the 60% overlap rule, applied exclusively to the accuracy distributions of the evaluated models. The analysis was limited to the accuracy metric, as extending the pairwise comparisons to all evaluation metrics would result in an excessively large and impractical table.

Model pairs exhibiting less than 60% overlap were identified as having a significant performance difference, enabling a robust and interpretable comparison of algorithms while accounting for variability across cross-validation folds.

Table 6 presents the pairwise significance testing results based on the accuracy distributions of the evaluated models using the 60% overlap rule. The analysis shows that a subset of model pairs, approximately 74% of all comparisons, exhibit statistically significant differences in performance, indicating meaningful distinctions and supporting the overall reliability of the results.

Exploratory Data Analysis and Feature Impact Assessment: In this section, a multistage exploratory analysis was conducted to gain a deeper understanding of the role of each feature in predicting the target variable. This analysis includes statistical feature–target association, model-based feature importance evaluation, and sensitivity analysis of model performance with respect to feature removal. In the first step, point-biserial correlation and Spearman's rank correlation were employed to quantify the relationship between individual features and the binary target variable. The results indicate that features such as Chest pain, Slope, and Maximum heart rate exhibit a significant positive association with the target, whereas Number of vessels, ST depression, and Exercise-induced angina show a significant negative correlation ($p < 0.01$). The consistency observed between the two correlation measures highlights the robustness of the identified relationships and suggests the presence of stable underlying patterns in the data.

Subsequently, feature importance was estimated using four different models, namely Random Forest, XGBoost, LightGBM, and Logistic Regression. To ensure comparability across models, the importance values were normalized, and their average was considered as the final model-based importance score. The results reveal that Chest pain consistently achieves the highest contribution across all models, obtaining the largest average importance weight. Additionally, features such as Slope, Thalassemia, and Number of vessels also demonstrate high importance across multiple models, indicating their critical role in the decision-making process. This observation suggests that despite differences in model architectures and underlying assumptions, certain features provide stable and highly informative predictive signals.

To assess the practical relevance of each feature on model performance, a feature ablation analysis based on the decrease in the F1-score was performed. In this analysis, removing the Chest pain feature resulted in the largest performance degradation ($\Delta F_1 \approx 0.04$), confirming its essential contribution to the predictive structure of the model. In contrast, the removal of features such as Age, Sex, and Serum cholesterol led to negligible changes in performance, which may indicate information redundancy or a secondary role of these variables in the presence of other features. These findings emphasize the importance of distinguishing between statistical relevance and performance-driven importance when interpreting feature contributions.

Finally, by integrating three complementary criteria—statistical correlation, model-based importance, and performance degradation due to feature removal—a unified final score was computed to rank the overall impact of the features. This combined analysis identifies Chest pain as the most influential feature by a substantial margin, followed by Slope, Number of vessels, and Thalassemia. Such a comprehensive, multi-perspective evaluation enhances the interpretability of the developed models and provides strong

Table 6 Pairwise accuracy-based significance testing between models

Algorithm 1	Algorithm 2	Significant difference
RF	XGB	No
RF	SVM	Yes
RF	LR	Yes
RF	MLP	No
RF	LGBM	No
RF	KNN	Yes
RF	DT	Yes
RF	ETC	No
RF	AB	Yes
RF	VE-NB	Yes
RF	LDA	Yes
RF	QDA	Yes
XGB	SVM	Yes
XGB	LR	Yes
XGB	MLP	No
XGB	LGBM	No
XGB	KNN	Yes
XGB	DT	No
XGB	ETC	No
XGB	AB	Yes
XGB	VE-NB	Yes
XGB	LDA	Yes
XGB	QDA	Yes
SVM	LR	Yes
SVM	MLP	Yes
SVM	LGBM	Yes
SVM	KNN	No
SVM	DT	No
SVM	ETC	Yes
SVM	AB	Yes
SVM	VE-NB	Yes
SVM	LDA	Yes
SVM	QDA	Yes
LR	MLP	Yes
LR	LGBM	Yes
LR	KNN	Yes
LR	DT	Yes
LR	ETC	Yes
LR	AB	Yes
LR	VE-NB	Yes
LR	LDA	No
LR	QDA	Yes
MLP	LGBM	No
MLP	KNN	Yes
MLP	DT	No
MLP	ETC	No
MLP	AB	Yes
MLP	VE-NB	Yes
MLP	LDA	Yes
MLP	QDA	Yes
LGBM	KNN	No
LGBM	DT	No
LGBM	ETC	No

Table 6 (continued)

Algorithm 1	Algorithm 2	Significant difference
LGBM	AB	Yes
LGBM	VE-NB	Yes
LGBM	LDA	Yes
LGBM	QDA	Yes
KNN	DT	No
KNN	ETC	No
KNN	AB	Yes
KNN	VE-NB	Yes
KNN	LDA	Yes
KNN	QDA	Yes
DT	ETC	No
DT	AB	Yes
DT	VE-NB	Yes
DT	LDA	Yes
DT	QDA	Yes
ETC	AB	Yes
ETC	VE-NB	Yes
ETC	LDA	Yes
ETC	QDA	Yes
AB	VE-NB	Yes
AB	LDA	Yes
AB	QDA	Yes
VE-NB	LDA	Yes
VE-NB	QDA	Yes
LDA	QDA	Yes

empirical support for the algorithmic comparisons and statistical significance tests presented in the earlier sections of this study.

Table 7 summarizes the performance of various machine learning models reported in previous studies on heart disease datasets, compared to the models developed in this study. Across multiple datasets, including the Cleveland Heart Disease dataset (303 patients, 14 features) and the larger Heart Disease dataset (1025 patients, 14 features), our models consistently achieved higher accuracy and precision than prior works. For instance, while previous studies reported accuracies in the range of 80–91% for Random Forest and XGBoost models, our approach attained accuracies of 93.82% for Random Forest and 93.17% for XGBoost, along with similarly high precision values (94.57% and 94.07%, respectively).

Moreover, the SVM, KNN, and Decision Tree algorithms in our study demonstrated superior performance compared to previous works. Other algorithms included in our study, which are not shown in this table, were not evaluated in prior studies due to their absence, highlighting the comprehensiveness of our work in examining a wide range of machine learning models for heart disease prediction.

Notably, the proposed ensemble model combining Voting and Naive Bayes (VE-NB) also demonstrated competitive performance, achieving 70.68% accuracy on more challenging models where individual algorithms struggled, highlighting the robustness of our methodology across different classifiers. Overall, the results indicate that the feature selection, model optimization, and integration strategies implemented in this study provide a clear improvement over prior approaches, both in terms of predictive accuracy and generalizability. These findings underscore the effectiveness of the proposed

Table 7 Comparison of machine learning model performance with previous studies

Authors	Dataset	Machine learning models						
		DT	KNN	LR	NB	RF	SVM	XGB
Ahmad et al. [1]	Heart Disease Dataset (1025 patients, 14 features), Heart Disease UCI Kaggle Dataset (918 patients, 12 features)	–	Acc: 88.77	**Acc: 88.04**	–	–	Acc: 88.41	Acc: 88.33
Singh et al. [4]	Heart Disease Dataset on UCI (303 patients, 14 features)	–	–	Acc: 85.20	–	Acc: 88.50	–	Acc: 83.60
Ramalingam et al. [5]	Heart Disease Dataset on UCI (303 patients, 14 features)	–	–	–	**Acc: 83.49**	Acc: 91.60	–	–
Patel et al. [7]	Heart Disease Dataset on UCI (303 patients, 14 features)	–	–	Acc: 87.31	–	Acc: 80	–	–
Srivenkatesh [34]	Cardiovascular Disease Dataset on Kaggle (1090 patients, 10 features)	–	Acc: 58.71	Acc: 77.06	Acc: 69.72	Acc: 74.31	Acc; 71.55	–
Rahim et al. [35]	Framingham Heart Study Dataset on Kaggle (4240 patient, 15 features)	Acc: 74.3	Acc: 83.4	Acc: 84.2	–	Acc: 82	Acc: 79	–
Ahamad et al. [36]	Heart Disease Cleveland (303 patients, 14 features), Heart Disease Dataset (1025 patients, 14 features)	Acc: 79.02	Acc: 85.68	–	–	–	**Acc: 91.33**	Acc: 89.08
Adil et al. [37]	Cardiovascular Disease Dataset (70,000 patients, 12 features)	Acc: 63 Pre: 64	–	–	Acc: 72 Pre76	Acc: 72 Pre: 72	Acc: 72 Pre: 78	Acc: 73 Pre: 70
Mounita et al. [38]	CLD Dataset UCI (583 patients, 10 features)	Acc: 79 Pre: 79	Acc: 81 Pre: 83	Acc: 79 Pre: 83	–	Acc: 84 Pre: 87	Acc: 81 Pre: 81	Acc: 82 Pre: 87
Our Article	Heart Disease Cleveland dataset Kaggle (303 patients, 14 features), Heart Disease Dataset (1025 patients, 14 features)	**Acc: 92.05 Pre: 92.13**	**Acc: 91.65 Pre: 92.63**	Acc: 73.65 Pre: 72.99	VE-NB: Acc: 70.68 Pre: 70.24	**Acc: 93.82 Pre: 94.57**	**Acc: 91.00 Pre: 91.70**	**Acc: 93.17 Pre: 94.07**

methods and confirm their potential utility for clinical decision support in cardiovascular disease diagnosis.

7 Conclusion and discussion

This study systematically evaluated the performance of a diverse set of machine learning algorithms for heart disease prediction, with particular emphasis on the role of key features in model decision-making. Tree-based models, notably Random Forest and Extra Trees, demonstrated superior accuracy, reliability, and stability compared to other approaches. These models effectively identified high-risk patients while minimizing diagnostic errors, underscoring their potential utility in clinical decision-support systems.

Gradient boosting algorithms also exhibited competitive and balanced performance in terms of sensitivity and discriminative ability, whereas linear and simpler models showed limited capacity to capture the complex, nonlinear patterns inherent in medical data. These findings highlight the critical importance of selecting models suited to the complexity of clinical datasets.

Feature impact analysis revealed that certain variables exert a central influence on predictive performance. Integrating statistical association, model-based importance, and ablation analysis provided a robust framework for assessing feature relevance, enhancing both interpretability and practical applicability of the models.

Comparison with previous studies indicated a notable improvement in predictive accuracy and robustness, demonstrating that feature selection, model optimization, and ensemble strategies can significantly enhance performance. Overall, these results confirm that tree-based and ensemble approaches, combined with rigorous feature analysis, offer reliable and interpretable tools for early detection and clinical decision-making in cardiovascular disease.

Although data integration was performed by merging two independent datasets, the size of the resulting integrated dataset remains relatively limited, which may still constrain the generalizability of the findings.

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Author contributions
M.E.A. is the sole author of the article and has done all the work on this article and checked all the files and steps.

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Data availability
The datasets generated and/or analysed during the current study are available in the Kaggle repository: https://www.kaggle.com/datasets/ritwikb3 /heart-disease-cleveland [14]. Heart Diseas Dataset on Kaggle: https://www.kaggle.com/datasets/johnsmit h88/heart-disease-dataset [45].

Declarations

Ethics approval
The study utilized two publicly available dataset; therefore, no ethical approval or consent was required.

Consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interest
The authors declare no competing interest.

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